

Self Learning LLM agents

Metareflection: Learning Instructions for Language Agents Using Past Reflections

Priyanshu Gupta*, **Shashank Kirtania***, Ananya Singha*, Sumit Gulwani, Arjun Radhakrishna, Sherry Shi, Gustavo Soares
[\[EMNLP 2024\]](#)

STACKFEED: Structured Textual Actor-Critic Knowledge Base Editing with FeedBack
Shashank Kirtania*, Naman Gupta*, Priyanshu Gupta, Sumit Gulwani, Arun Iyer, Suresh Parthasarathy, Arjun Radhakrishna, Sriram K Rajamani, Gustavo Soares
[\[DL4Code NeurIPS 2025\]](#)
[\[EMNLP 2025\]](#)

Improving Language Agents Through BREW
Shashank Kirtania, Param Biyani, Priyanshu Gupta, Yasharth Bajpai, Roshni Iyer, Sumit Gulwani, Gustavo Soares
[\[MTI Workshop at NeurIPS 2025\]](#)
[\[COLM 2026\]](#)

Logic-Im++: Multi-step Refinement for Symbolic Formulations
Shashank Kirtania, Priyanshu Gupta, Arjun Radhakrishna
[\[NLRSE workshop at ACL 2024\]](#)

Formal Reasoning with LLMs

Activation Steering in Neural Theorem Provers
Shashank Kirtania, Arun Iyer
[\[EM in Wild at ICLR 2025\]](#)

IndiMathBench: Autoformalizing Mathematical Reasoning Problems with a Human Touch
Param Biyani, **Shashank Kirtania**, Yasharth Bajpai, Sumit Gulwani, Ashish Tiwari
[\[FM-AI workshop at AAAI 2026\]](#)
[\[Under review\]](#)

"Code is cheap, show me the specification": Training LLMs for end-to-end Verified Code Generation
Swayam Singh*, Ramneet Singh*, **Shashank Kirtania***, Tianyu Chen, Aditya Kanade, Shan Lu, Nagarajan Natarajan
[\[Under review\]](#)

AI for Software Engineering